

FIFA Data Structures and Search Techniques

DESCRIPTION:

I HAVE TAKEN A FIFA WORLD CUP DATASET FROM 1930 TO 2018. PARSED IT AND IMPLEMENTED DATA STRUCTURE TECHNIQUES TO ENSURE LEAST MEMORY USAGE AND EFFICIENT SEARCHING AND INSERTION OF DATA.

STRUCTURES USED:

#	Data Structure	Purpose
1	AVL TREES	TO STORE DATA HENCE RESULTS IN FASTER SEARCHING.
2	HASH TABLES (ARRAY)	GIVES INSTANT ACCESS TO VALUE USING HASH FUNCTION
3	SINGLY LINKED LIST	STORE ITEMS NOT USED IN SEARCHING. IT PROVIDES EASY TRAVERSAL.

GOALS:

- Achieving a memory efficient data structure by storing data at one place and using pointers to point to that data when required.
- Use of AVL Trees for better searching functions and better management of data.
- Using hash tables for storing linked list and storing the pointers to the actual data of AVL trees in the nodes of linked list ensuring quick access to the data and less memory consumption.
- Use of linked list for easy traversal of data
- Less focus on creating methods and more focus on data structuring and managing techniques.

EXPLANATION:

- Match.h declares a class containing all the attributes of a match.
- Matchnode.h declares a class for a node containing data from match.h, lchild, rchild.

- Matchlist.h declares a class for an avl tree and some functions to insert the matchnode from parser
- Into an avl tree. Also, I have made a function that search head to head matches between two teams as an example of how to use this data structure.
- Matchtablenode.h declares a class for a node containing two pointers. One point to the objects already stored in the avl trees and one point to the next node.
- Matchtablelist.h declares a class for a singly linked list. The matchtablenodes will be stored in the singly linked list. One list will contain nodes representing matches of that specific world cup match.
- Worldcupyear.h declares a class for a hash table with data type as matchtablelist as it will store linked list at each array index.
- Parser.h which parses the data from the csv file and store it in vector from where I am receiving it as nodes of avl tree. One line per node. All the class objects are created in this class for functioning.
- Main.cpp just calls functions from parser1 class and run the functionalities defined in all the included classes.

RUNNING THE PROGRAM:

RUNNING THE CODE WILL ASK FOR 3 DIFFERENT OPTIONS FROM THE USER:

1. DISPLAY ALL FIFA MATCHES FROM 1930 – 2018.
2. FINDING DETAILS OF MATCHES BETWEEN TWO SPECIFIC COUNTRIES.
3. FINDING MATCHES OF A PARTICULAR WORLDCUP.

(OPTION 1 AND 2 ARE DONE USING SEARCH METHOD DEFINED ON AVL TREES AND OPTION 3 IS DONE USING LINKED LIST STORED IN HASH TABLES. ONCE AGAIN THE FOCUS WAS MORE ON STRUCTURING THE DATA, WE CAN CREATE METHODS EASILY ONCE DATA IS MANAGED)

Option when running the code:

DSA PROJECT: FIFA MATCHES DATA STRUCTURING

SELECT THE OPTIONS:

1. PRINT ALL THE MATCHES FROM 1930-2018.
2. CHECK HEAD TO HEAD MATCHES BETWEEN TWO COUNTRIES
3. CHECK MATCHES OF A PARTICULAR WORLD CUP

OPTION 1:

```
895 Saturday Sweden 0 England 2 Russia Russia 2018 Jul Samara Sweden England A Quarterfinals
896 Saturday Russia 2 Croatia 2 Russia Russia 2018 Jul Sochi Russia Croatia Croatia won in penalties (4 - 3) A Quarterfinals
897 Tuesday France 1 Belgium 0 Russia Russia 2018 Jul Saint Petersburg Belgium France H Semifinals
898 Wednesday Croatia 2 England 1 Russia Russia 2018 Jul Moscow England Croatia Croatia won in AET H Semifinals
899 Saturday Belgium 2 England 0 Russia Russia 2018 Jul Saint Petersburg England Belgium H Third place
900 Sunday France 4 Croatia 2 Russia Russia 2018 Jul Moscow Croatia France H Final
```

OPTION 2:

```
2
ENTER TWO COUNTRIES: (format : Argentina)
Argentina

Brazil

Argentina 1 2 Brazil Group A 1974
Argentina 0 0 Brazil Group B 1978
Argentina 1 3 Brazil Group C 1982
Brazil 0 1 Argentina Round of 16 1990
```

OPTION 3:

ENTER THE WORLD CUP DATE:

1930

Sunday France 4 Mexico 1 1930 Group 1

Sunday Belgium 0 United States 3 1930 Group 4

Monday Brazil 1 Yugoslavia 2 1930 Group 2

Monday Peru 1 Romania 3 1930 Group 3

Tuesday Argentina 1 France 0 1930 Group 1

Wednesday Chile 3 Mexico 0 1930 Group 1

Thursday Bolivia 0 Yugoslavia 4 1930 Group 2